

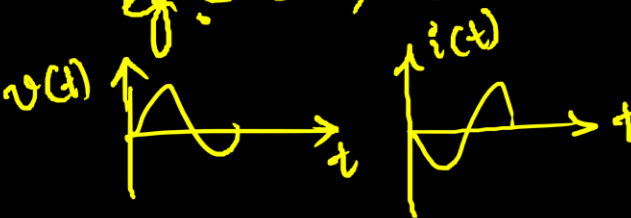
ANALOG CHARACTERIZATION



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B.Tech (ECE | Silicon University)
Specialization in Microelectronics & VLSI

In CHARACTERIZATION; We'll
Basically deal with 3 types of
Inputs $\rightarrow$ Namely :-

i. Any HARMONIC function
eg:- sin, cosine


ii. PULSE/RECTANGULAR/COSVARE


$\rightarrow$ PWM, PCM, DM $\rightarrow$ Communication
- step size
- Encoding/Quantization

ANALOG CHARACTERIZATION :-

Continuous time
domain signal

A-C signal : $v(t)$
small signal
 $\downarrow$
ampl.

$$i(t)/I(t) \\ \downarrow \\ i(t) = \frac{dq}{dt} \\ q = ne$$

D-C : $V(t)$
signal

t : is always an
Independent
variable =

Dynamic
variable =

characteristic of a DEVICE $\rightarrow$ I. ACTIVE T_r
+ II. PASSIVE R, C
 $\downarrow$
Improvisation/modifications
using certain parameters, E_s 's, N_{ox} , etc.

DEVICE PARAMETERS

MODELS

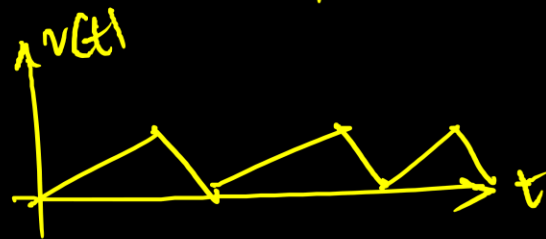
eg $\rightarrow$ BSIM

Berkely Semulⁿ
Model.

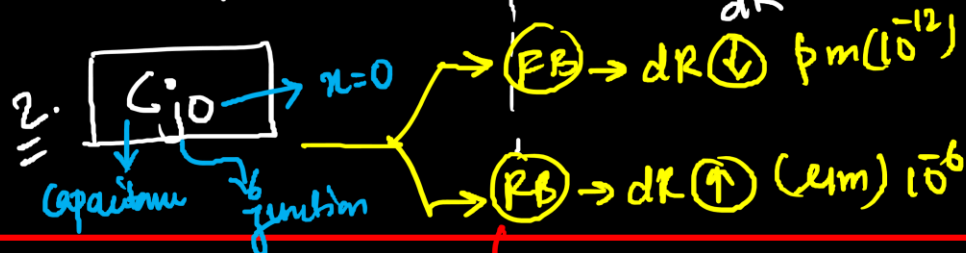
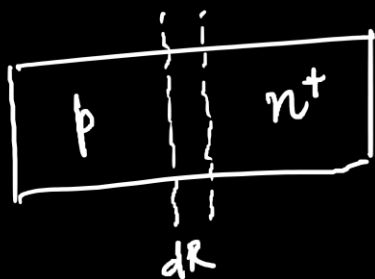
" MOSFET "

$\rightarrow$ SPICE Models $\rightarrow$ I
II
III

iii. SAW TOOTH / TRIANGULAR



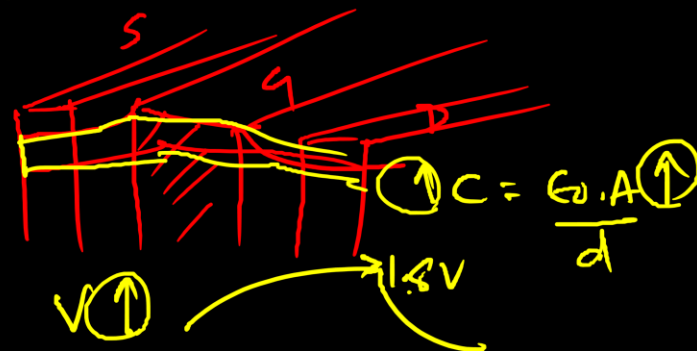
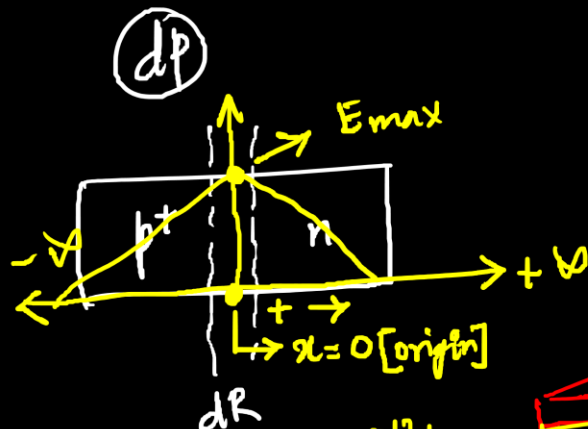
(dn)



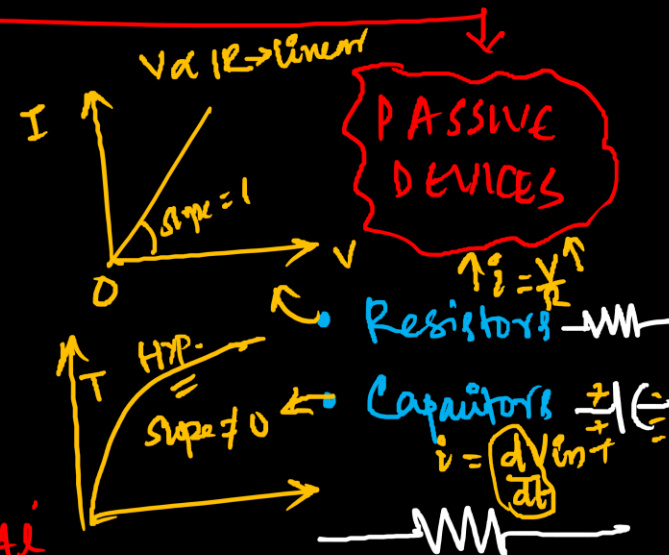
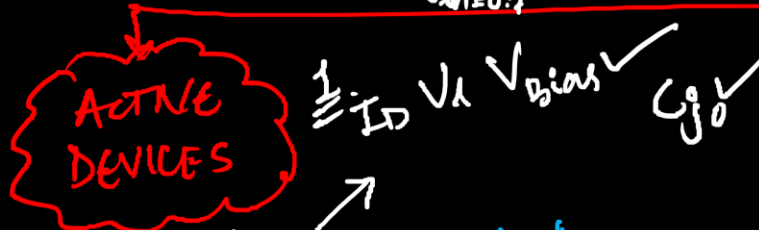
"TRIODE"

- Diodes $\rightarrow$ 2 terminal device
- Transistors [Bipolars] $\rightarrow$ 3 terminal device
- MOSFETS $\rightarrow$ 4/5 terminal device

Pentode



CHARACTERIZATION



Heat resistant material $\rightarrow$ poly $\rightarrow$ Material

Conductivity $\uparrow \uparrow$

Cadence $\rightarrow$ Model $\rightarrow$ pres [poly-resistor].

MIM, SIM, opp, cmu3, cmu4

PMOS, NMOS
= CMOS

MOSFETS → "CMOS Inverter"

✓ 1. I_D vs V_{GS} → Threshold voltage → V_{th}

✓ 2. I_D vs V_{DS} → λ (Chm coefficient) = $\lambda = \frac{1}{r_o} = \frac{1}{V_A} = \frac{1}{\text{Early Voltage}}$
op Intaintz Resistance =

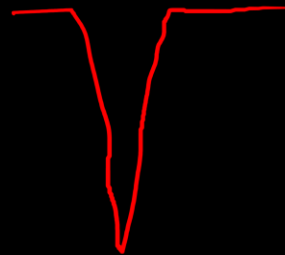
✓ 3. $\frac{\partial I_D}{\partial V_{GS}} \rightarrow \bar{g}_m$; Transconductance [i/p]

✓ 4. $\frac{\partial I_D}{\partial V_{DS}} \rightarrow \bar{g}_{DS}$; Transconductance [o/p]

5. $\frac{\partial V_{out}}{\partial V_{in}}$: Noise Margins/Immunity/Tolerance

6. Process Parameter → $\mu_n C_{ox}$: K : Kappa.
✓ = $\frac{C_{ox}}{t_{ox}}$

• $NMH = V_{OH} - V_{IH}$
• $NML = V_{IL} - V_{OL}$



7. Propagation Delay = o/p - i/p.

8. Static Power Consumption, ϕ Dynamic Power Consumption.

9. Energy delay product (EDP)

10. Switching Threshold -

11. $W_n = W_p$ ✓ → dc sweep Analysis
 $W_p > W_n$ ✓
 $W_p < W_n$ ✓ } if ch should be const.

ideally; $W_p = 2.5 \times W_n$.

12. Body Bias Effect

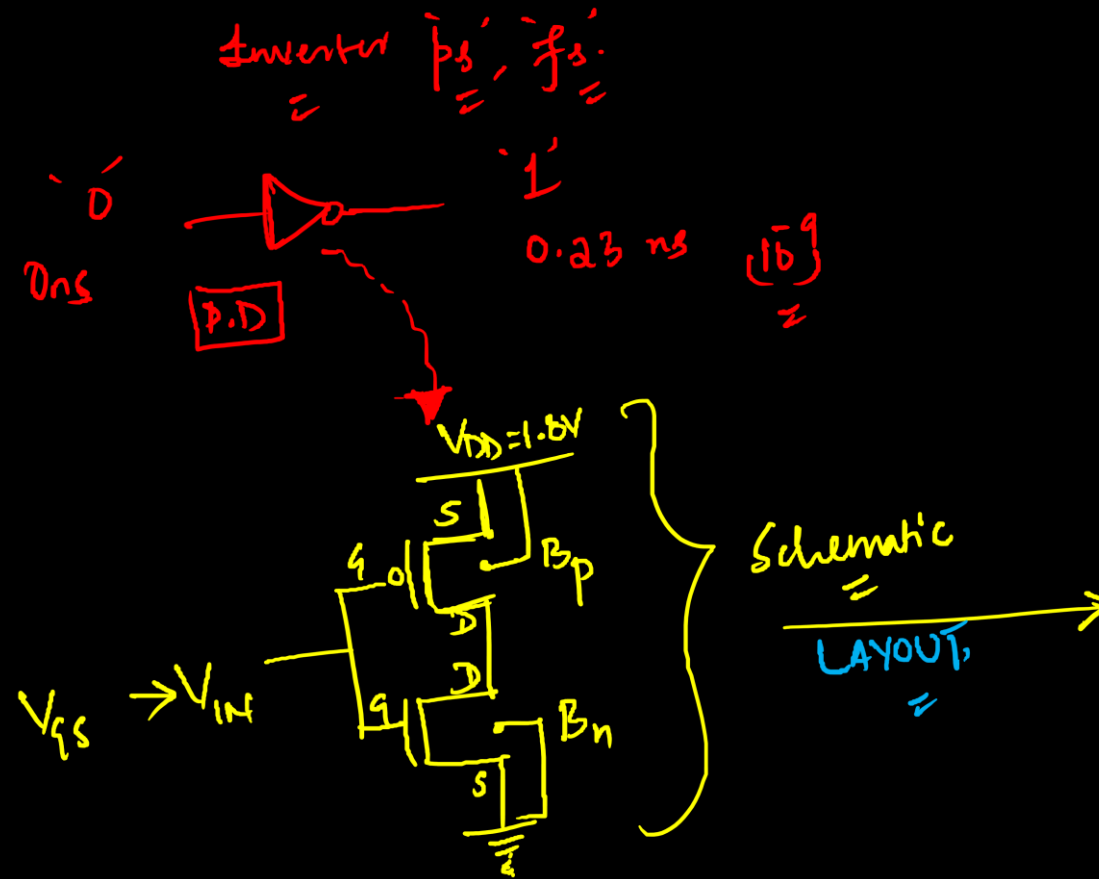
13. g_m vs R_{out}

14. $\log I_D$ Vs g_m

15. R_{on} Vs W_p or W_n

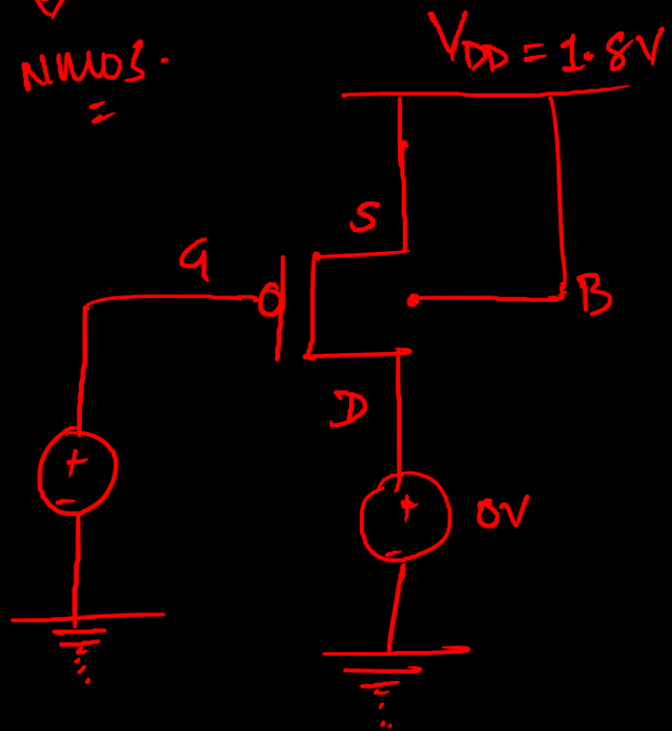
16. Delay Vs $P_{consumption}$

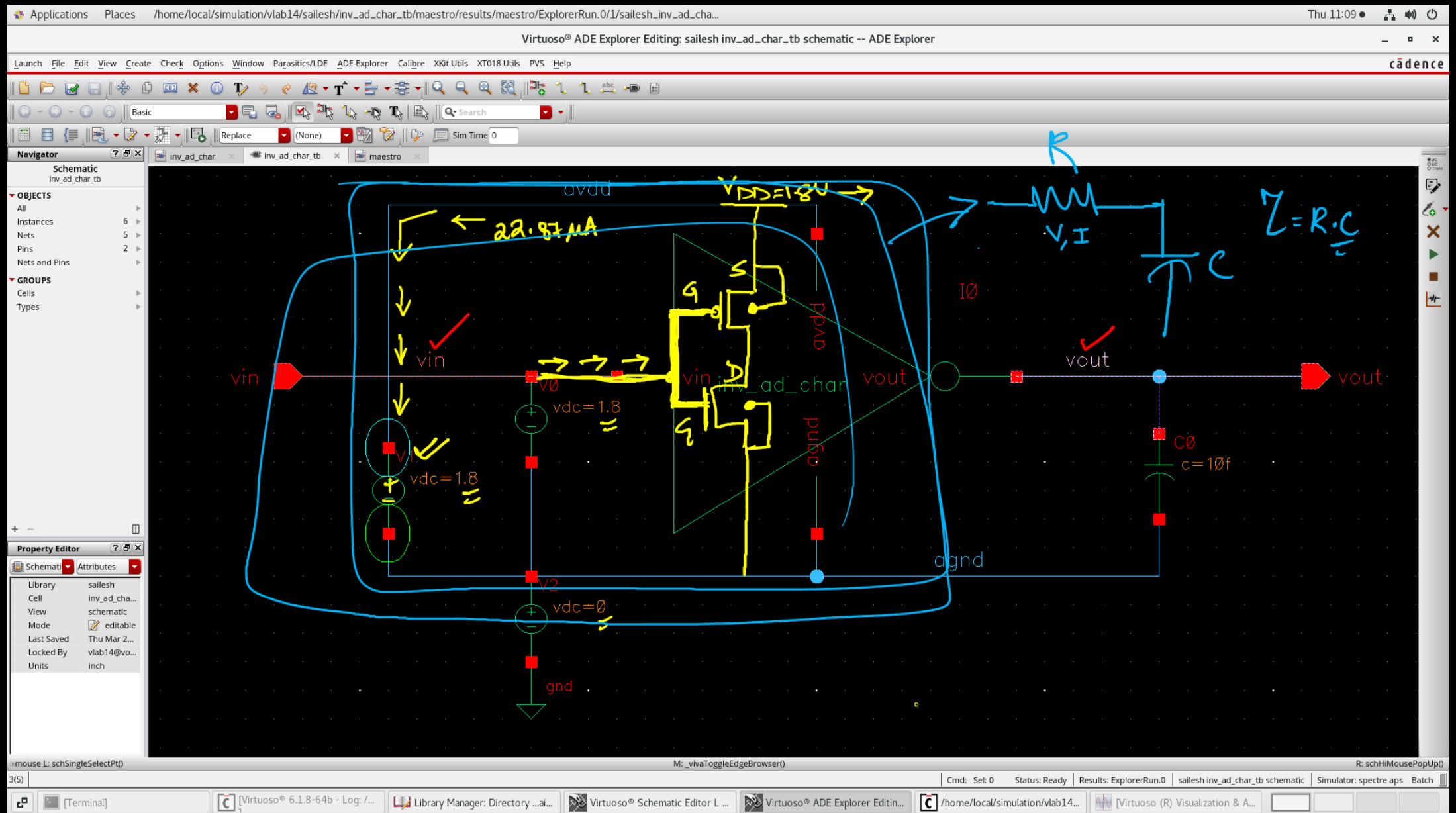
17.



I_D V_S V_{GS} :-

$\downarrow$
PMOS
 $\downarrow$
NMOS





Applications Places /home/local/simulation/vlab14/sailesh/inv_ad_char_tb/maestro/results/maestro/ExplorerRun.0/1/sailesh_inv_ad_cha... Thu 11:09

Virtuoso® ADE Explorer Editing: sailesh_inv_ad_char_tb maestro

Launch Session Setup Analyses Variables Outputs Simulation Results Tools EAD Parasitics/LDE Window Help

Replace (None)

= ANALOG DESIGN ENVIRONMENT =

Setup

Name	Type	Details	Value	Plot	Save	Spec
sailesh_inv_ad_char_tb_1	signal	/vout		☒	☐	
Simulator spectre	signal	/vin		☒	☐	
Analyses	signal (I)	/V1/MINUS		☒	☐	
dc 0 1.8 Automatic Start-Stop /V0	signal (I)	/V1/PLUS		☒	☐	

Filter

sailesh_inv_ad_char_tb_1

Simulator spectre

Analyses

dc 0 1.8 Automatic Start-Stop /V0

Click to add analysis

Design Variables

Click to add variable

Parameters

Corners

Reliability Analyses

Monte Carlo Sampling

Checks/Asserts

Choosing Analyses -- ADE Explorer

Analysis

tran ☒ dc ☐ ac ☐ noise

xf ☐ sens ☐ dcmatch ☐ acmatch

stb ☐ pz ☐ lf ☐ sp

envlp ☐ pss ☐ pac ☐ pstb

pnoise ☐ pxf ☐ psp ☐ qpss

qpac ☐ qpnoise ☐ qpaxf ☐ qpssp

hb ☐ hbac ☐ hbstb ☐ hbnoise

hbsp ☐ hbxf

DC Analysis

Save DC Operating Point ☐

Hysteresis Sweep ☐

Sweep Variable

Temperature ☐

Design Variable ☐

Component Parameter ☒

Model Parameter ☐

Component Name /V0

Select Component

Parameter Name dc

Sweep Range

Start-Stop ☒ Start 0 Stop 1.8

Center-Span ☐

Sweep Type

Automatic ☒

Add Specific Points ☐

Add Points By File ☐

Enabled ☒

Options...

OK Cancel Defaults Apply Help

Results: ExplorerRun.0 sailesh_inv_ad_char_tb schematic Simulator: spectre aps Batch

Terminal

Virtuoso® 6.1.8-64b - L...

Library Manager: Director...

Virtuoso® Schematic Edit...

Virtuoso® ADE Explorer ...

/home/local/simulation/vL...

Virtuoso (R) Visualizatio...

Choosing Analyses -- AD...

```
Applications Places /home/local/simulation/vlab14/sailesh/inv_ad_char_tb/maestro/results/maestro/ExplorerRun.0/1/sailesh_inv_ad_cha... Thu 11:09 ● [Icons] [Volume] [Power]

/home/local/simulation/vlab14/sailesh/inv_ad_char_tb/maestro/results/maestro/ExplorerRun.0/1/sailesh_inv_ad_char_tb_1/psf/spectre.out - [Close] [Maximize] [Fullscreen]

File Edit View Help cadence

Spectre (R) Circuit Simulator
Version 21.1.0.208.isr3 64bit -- 27 Nov 2021
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Includes RSA BSAFE(R) Cryptographic or Security Protocol Software from RSA Security, Inc.

User: vlab14 Host: voltadt03.vlsi.silicon.ac.in HostID: A8C0210B PID: 8245
Memory available: 1.7938 GB physical: 3.7876 GB
Linux : CentOS Linux release 7.9.2009 (Core)
CPU Type: Intel(R) Core(TM) i3-3240 CPU @ 3.40GHz
All processors running at 3400.0 MHz
Socket: Processors (Hyperthreaded Processor)
0: 0 ( 2 ), 1 ( 3 )

System load averages (1min, 5min, 15min) : 7.2 %, 5.8 %, 4.5 %
Hyperthreading is enabled
HPC is enabled

Simulating 'input.scs' on voltadt03.vlsi.silicon.ac.in at 11:07:28 AM, Thur Mar 27, 2025 (process id: 8245).
Current working directory: /home/local/simulation/vlab14/sailesh/inv_ad_char_tb/maestro/results/maestro/ExplorerRun.0/1/sailesh_inv_ad_char_tb_1/netlist
Command line:
/CAD/cadence/SPECTRE211/tools.lnx86/bin/spectre -64 input.scs \
+escchars +log ../psf/spectre.out -format psf1 -raw ../psf \
+aps +ltimeout 900 -maxw 5 -maxn 5 -env ade -ahdllibdir \
/home/local/simulation/vlab14/sailesh/inv_ad_char_tb/maestro/results/maestro/ExplorerRun.0/sharedData/CDS/ahd1/input.ahd1SimDB \
+logstatus

Licensing Information:
[11:07:29.502690] Configured Lic search path (20.02-p002): 5280@cdslicsrv.vlsi.silicon.ac.in:1717@mgclicsrv.vlsi.silicon.ac.in

Licensing Information:
[11:07:29.105760] Periodic Lic check successful

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Loading /CAD/cadence/SPECTRE211/tools.lnx86/cmi/lib/64bit/5.0/libstmodels_sh.so ...
Reading file: /home/local/simulation/vlab14/sailesh/inv_ad_char_tb/maestro/results/maestro/ExplorerRun.0/1/sailesh_inv_ad_char_tb_1/netlist/input.scs
Reading file: /CAD/cadence/SPECTRE211/tools.lnx86/spectre/etc/configs/spectre.cfg
Reading file: /PDK/xfab/XKIT/xt018/cadence/v12_0/spectre/v12_0_1/ip5mos/config.scs
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8 >
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